Longest Alternating Subsequence

Problem Statement

Given a number sequence, find the length of its Longest Alternating Subsequence (LAS). A subsequence is considered alternating if its elements are in alternating order.

A three element sequence (a1, a2, a3) will be an alternating sequence if its elements hold one of the following conditions:

```
{a1 > a2 < a3 } or { a1 < a2 > a3}.
```

Example 1:

```
Input: {1,2,3,4}
Output: 2
Explanation: There are many LAS: {1,2}, {3,4}, {1,3}, {1,4}
```

Example 2:

```
Input: {3,2,1,4}
Output: 3
Explanation: The LAS are {3,2,4} and {2,1,4}.
```

Example 3:

```
Input: {1,3,2,4}
Output: 4
Explanation: The LAS is {1,3,2,4}.
```

Try it yourself

Try solving this question here:

Java

```
1
2  class Solution {
3
4    public int findLASLength(int[] nums) {
5      // TODO: Write your code here
6      return -1;
```

```
8    }
9
```

Test Case | Run Code Result | Stdout

```
[1,2,3,4]
```

Use Example Testcases | Run | Submit

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